

Introduction to C Language

1. What is Programming?

- **Programming** is the process of writing a set of instructions that tells the computer how to perform a specific task.
- A **Program** is a collection of instructions written in a programming language.
- Computers can only understand **machine language** (binary: 0s and 1s).
- Humans cannot write programs directly in binary, so we use **programming languages** that are later converted into machine code.

2. Why Do We Need Programming?

- A computer is a **general-purpose machine** — it can do many things, but it needs clear instructions (programs) to perform any task.
- Without programs, the computer cannot work on its own.
- **Software** = A collection of programs.
 - **System Software**: Manages hardware (Example: Operating Systems like Windows, Linux).
 - **Application Software**: Helps users do tasks (Example: MS Office, Web Browsers, Games).

Main Reason: Programming allows us to solve real-world problems and automate tasks using computers.

3. Types of Programming Languages

- **Low-Level Languages:**
 - Machine Language (Binary – 0s and 1s)
 - Assembly Language (Uses short codes called mnemonics)
- **High-Level Languages:**
 - Easy for humans to understand and write (Examples: C, C++, Java, Python).
- High-level programs are converted into machine code using a **Compiler** or **Interpreter**.

4. Introduction to C Language

- **Developed by:** Dennis Ritchie
- **Year:** 1972
- **Developed at:** Bell Laboratories (AT&T), USA
- C was created as a successor to the **B language**.
- **Main Purpose:** To develop the **UNIX operating system**.

C is known as a **System Programming Language** because it is very close to the hardware and gives good control over system resources.

5. Importance and Uses of C Language

- Many important softwares and system components are written in C.
- Used in:
 - Operating Systems (Linux kernel, parts of Windows and UNIX)
 - Device Drivers
 - Compilers and Interpreters
 - Embedded Systems
 - Databases
 - Game development (performance-critical parts)
- C is fast, efficient, and portable (code can run on different machines with little change).
- In 1989, C was standardized by **ANSI** (American National Standards Institute) → known as **ANSI C** or **C89**.
- Many modern languages (C++, Java, etc.) are based on the syntax and features of C.
- C is often called the "**Mother of Modern Programming Languages**".

6. Key Characteristics of C

- Simple and small language (only 32 keywords in original C).
- High performance and speed.
- Supports structured programming.
- Gives direct access to memory (using pointers).
- Highly portable.

Summary / Key Takeaways

- Programming means giving instructions to the computer.
- We need programs because computers cannot think or act without instructions.
- C is a powerful, fast, and widely used system programming language.
- C was created by Dennis Ritchie in 1972 at Bell Labs primarily for developing UNIX.
- Learning C gives a strong foundation for understanding how computers and other languages work.